INTRODUCTION TO MACHINE LEARNING · TEAM 8

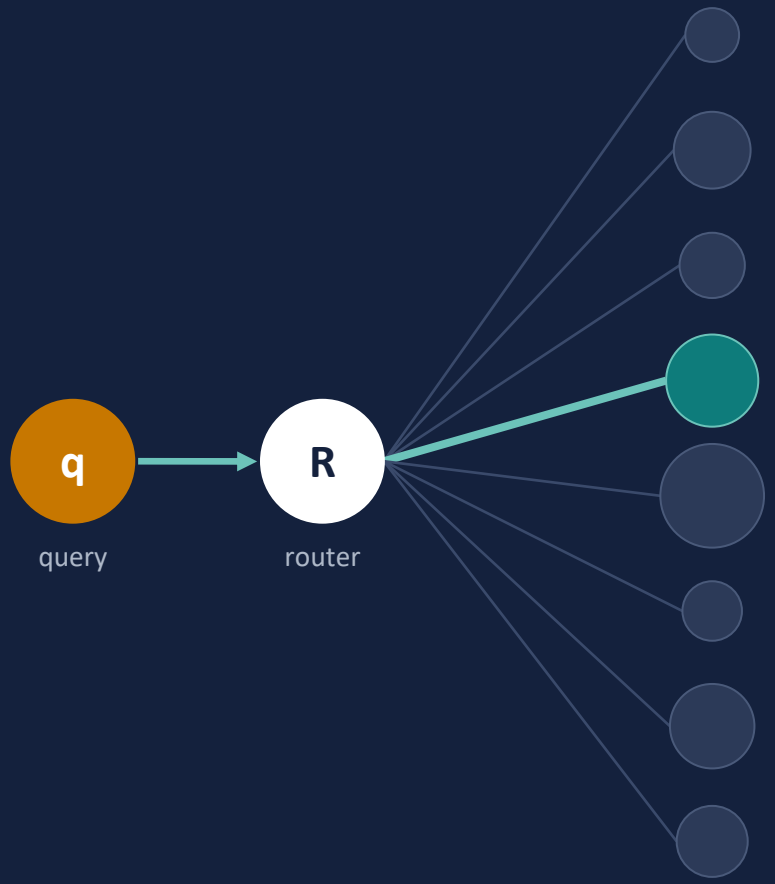
PILOT: Adaptive LLM Routing under Budget Constraints

Reading the paper and reproducing its two core experiments

Paper [1]: P. Panda, R. Magazine, C. Devaguptapu, S. Takemori, V. Sharma — Findings of EMNLP 2025

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24KHMT1 · University of Science, VNU-HCM · October 2026



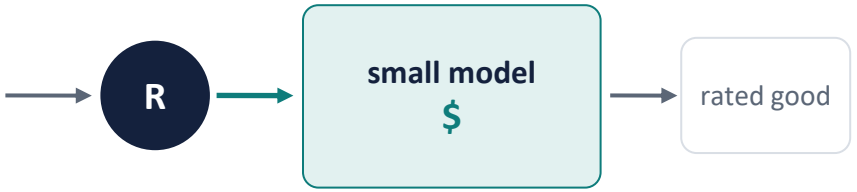
8 LLM arms (circle size ≈ cost)

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.

One query, eight models, one monthly budget

Our example · an HCMUS student help-desk chatbot

"When does course registration close?"



"Why does gradient descent diverge when the step size is too large? Show the math."



Only the answer that was shown gets a thumbs-up or thumbs-down. The other seven models stay unobserved.

1 Quality
The best model differs per query, and no one labels queries in advance.

2 Cost
Prices differ by more than 10×. The budget B covers the whole month, not one query.

3 Feedback
Bandit feedback: a reward only for the chosen model.

Goal maximize $\Sigma_t r_t$ subject to $\Sigma_t c_t(a_t) \leq B$, observing r_t only for the chosen model a_t [1]

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.

Two predecessors, two missing pieces

Approach	Core idea	Limitation left for LLM routing	Practical consequence
LinUCB [2] WWW 2010	Per-arm linear reward model fitted by ridge regression, plus an upper-confidence exploration bonus. Learns from the chosen arm only.	Every arm starts from zero (cold start). Optimizes reward with no budget.	Early users pay for exploration. Spend cannot be capped.
Hybrid LLM [3] ICLR 2024	Supervised classifier predicts if a small model is “good enough” versus a large one. A threshold sets the cost–quality trade-off.	Needs both models’ answers and quality scores for every training query. Two models only. Static after training.	Labeling cost grows with the pool. No adaptation from user feedback.
PILOT [1] EMNLP 2025	Preference prior + LinUCB-style updates + online budget policy, for any number of models.	New assumptions: the prior is close, cosine scores carry reward differences, cost estimates hold.	**These are the weaknesses our poster tests.**

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[2] L. Li, W. Chu, J. Langford, and R. E. Schapire, “A contextual-bandit approach to personalized news article recommendation,” in Proc. 19th Int. Conf. World Wide Web, 2010, pp. 661–670.
[3] D. Ding et al., “Hybrid LLM: Cost-efficient and quality-aware query routing,” in Proc. Int. Conf. Learn. Represent. (ICLR), 2024.

What the paper claims, grouped by kind

In the paper's terms

(i) LLM routing as a budget-constrained contextual bandit

(ii) PILOT: preference-prior LinUCB + online cost policy, with a lower regret bound

(iii) Beats bandit baselines: lowest regret, highest performance across budgets

1 Theoretical

A preference prior closer to the true reward vector than the origin gives a smaller regret upper bound (PI-OFUL, Prop. 2.1).

Not attempted before: Bandit regret analyses with an informed, rather than zero, starting point for routing.

2 Technical

Two-phase preference pretraining → prior-initialized LinUCB → ZCL online knapsack policy with bins and rollover.

Not attempted before: Routing many LLMs from thumbs-up/down feedback alone while honoring a per-period budget.

3 Empirical

Online learning/deployment protocol on RouterBench. 93% of GPT-4 performance at 25% of its cost (MMLU: 86% at 27%).

Not attempted before: An online bandit-feedback benchmark protocol for LLM routing.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.

Notation and the formal problem

L, K	set of LLM arms, and how many (paper 11, ours 8)
q_t	query at round t
$\varphi(q) \in \mathbb{R}^D$	frozen embedding (text-embedding-3-small, D = 1,536)
$\psi(q) = W\varphi(q) + b \in \mathbb{R}^d$	learned projection (ours: d = 64)
$x_t = \psi(q_t)/\|\psi(q_t)\|$	unit context seen by the bandit
$\theta_a, \theta_a^{pref}$	arm vector, and its preference prior
$r_t \in [0, 1]$	reward of the chosen arm only
$c_t(a), \hat{c}_t(a)$	true and estimated cost of arm a on q_t
$A_a, b_a, \tilde{\theta}_a = A_a^{-1}b_a$	ridge statistics and current estimate
λ_a, α	prior strength, exploration weight

The problem

for $t = 1, \ldots, Q$: choose $a_t \in L$

maximize $\Sigma_t r_t$

subject to $\Sigma_t c_t(a_t) \leq B$

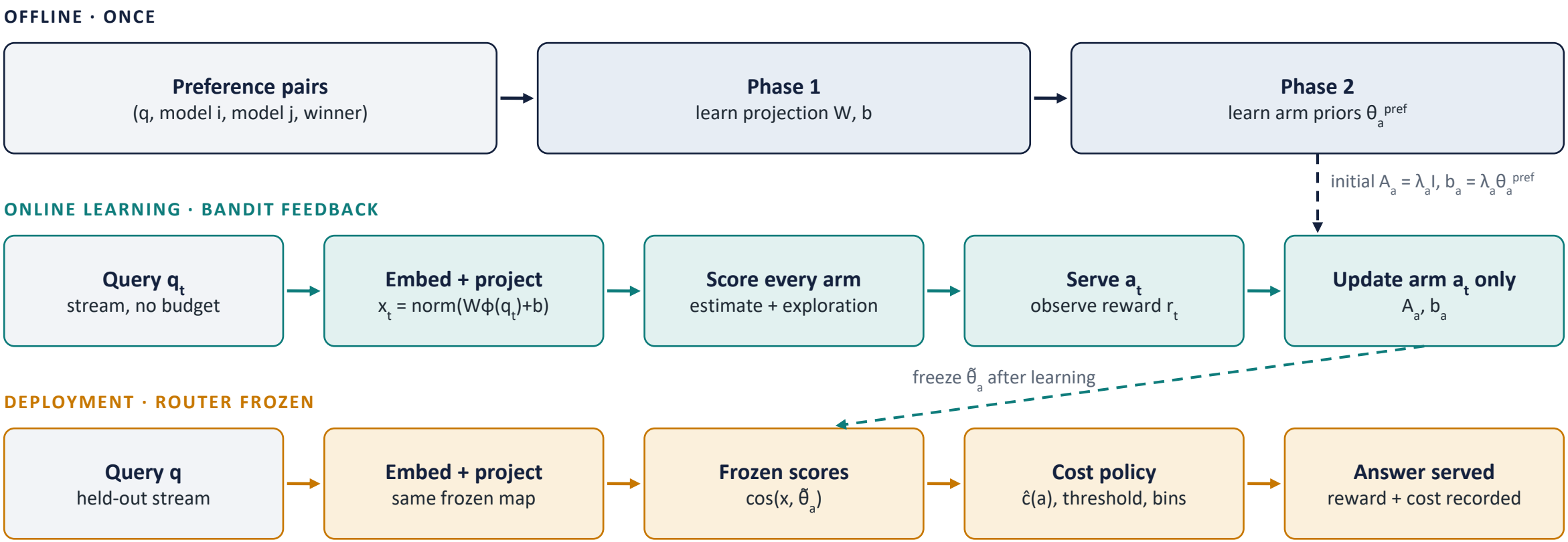
observe only $r_t = s(q_t, y_t(a_t))$

$y_t(a_t)$ is the chosen model's answer and s is a quality scorer. Success is measured by cumulative regret while learning and by reward on all requested queries, relative to GPT-4, when deployed.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.

METHOD · LAYER 2 OF 5

Pipeline: pretrain once, learn online, deploy frozen



Our run ϕ = text-embedding-3-small [5], real 1,536-d full-text vectors → learned 64-d · 8 of 11 arms · replayed outcomes, no LLM calls

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[4] W.-L. Chiang et al., "Chatbot Arena: An open platform for evaluating LLMs by human preference," arXiv:2403.04132, 2024.
[5] OpenAI, "New embedding models and API updates," OpenAI, Jan. 25, 2024. [Online]. Available: openai.com/index/new-embedding-models-and-api-updates

Pretrain a shared space from human preferences

What A space where cos(query, LLM) predicts how well that LLM fits that query.

Why Bandits that start from zero explore expensively. Human preference data already exists.

How Two phases, so the query map and the arm vectors do not chase each other.

Phase 1 · project the queries

$$\psi(q) = W\,\varphi(q) + b, \quad d(u, v) = 1 - \cos(u, v)$$

$$\ell = \max(0, d(\psi(q_a), \psi(q^+)) - d(\psi(q_a), \psi(q^-)) + m)$$

W, b learned map from $\mathbb{R}^D$ to $\mathbb{R}^d$
q^+ a query won by the same model as q_a
q^- hard negative: that model lost to a cheaper one
m margin (ours 0.2)

Phase 2 · one vector per LLM (W, b frozen)

$$p_i = \exp \cos(\theta_i, \psi(q)) \,/\, \Sigma_k \exp \cos(\theta_k, \psi(q)), \quad k \in \{i, j\}$$

$$\ell = \mathrm{BCE}(p_i, 1[\text{model i won}])$$

θ_i vector of LLM i in the shared space
p_i probability that model i beats model j on q
θ_i^{pref} result after training = the prior
BCE binary cross-entropy against the human label

Our run text-embedding-3-small [5] 1,536-d → d = 64 · 1,245 pairs → 1,193 triplets · 30 + 40 epochs · calibration 186 / 190 / 194 of 313 (60.70%)

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[4] W.-L. Chiang et al., “Chatbot Arena: An open platform for evaluating LLMs by human preference,” arXiv:2403.04132, 2024.
[5] OpenAI, “New embedding models and API updates,” OpenAI, Jan. 25, 2024. [Online]. Available: openai.com/index/new-embedding-models-and-api-updates

Learn online from one reward per query

1 **Reward model**

$$E[r \mid a, x_t] = \cos(x_t, \theta_a)$$

2 **Start at the prior**

$$A_a \leftarrow \lambda_a I, \quad b_a \leftarrow \lambda_a \theta_a^{\text{pref}} \quad \Rightarrow \quad \tilde{\theta}_a = \theta_a^{\text{pref}}$$

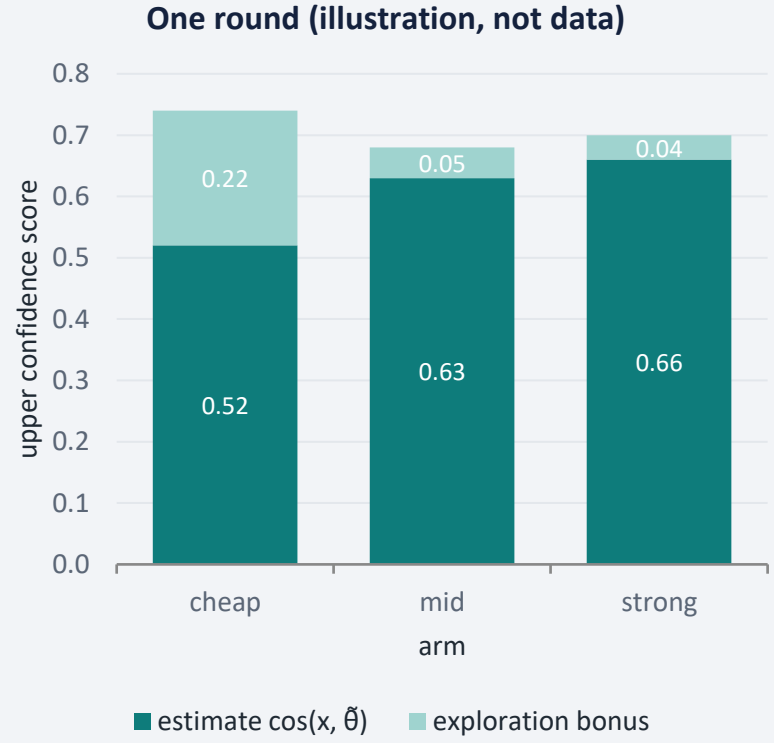
3 **Choose optimistically**

$$a_t = \operatorname{argmax}_a \cos(x_t, \tilde{\theta}_a) + \alpha\sqrt{(x_t^\top A_a^{-1} x_t)}$$

4 **Update the chosen arm only**

$$\text{for } a = a_t: \quad A_a \leftarrow A_a + x_t x_t^\top, \quad b_a \leftarrow b_a + r_t x_t$$

λ_a = 1 / (calibration preference accuracy of arm a) is a heuristic. α is tuned on the tuning split (ours: α = 1). No budget applies while learning.



The least-tried arm wins on its bonus (0.74), is served, and its bonus shrinks next time.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[2] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in Proc. 19th Int. Conf. World Wide Web, 2010, pp. 661–670.

Spend the budget online, bin by bin

1 **Estimate cost before answering**

$$\hat{c}_t(a) = \text{input tokens} \times \text{price}_{in} + \text{mean output tokens}_{tuning} \times \text{price}_{out}$$

2 **Eligibility shrinks as the bin fills**

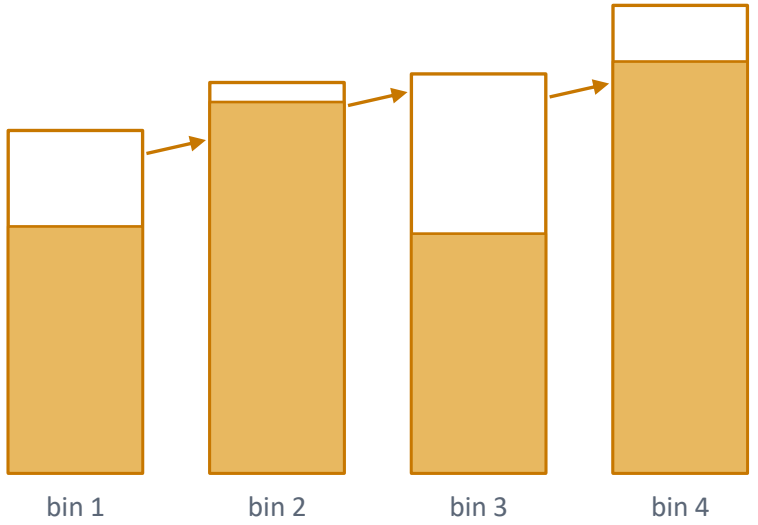
$$\hat{c}_t(a) \leq \cos(x_t, \tilde{\theta}_a) \,/\, ((UB \cdot e/LB)^z \cdot (LB/e)), \quad z = \text{spent} \,/\, \text{bin budget}$$

3 **Serve the best eligible arm, else fall back, else stop**

$$\text{guard } \hat{c}_t(a) \leq \text{money in bin} \cdot \text{fallback } \hat{c}_t(a) \leq B_{left} \,/\, Q_{left} \cdot \text{none} \Rightarrow \text{stop}$$

Guarantee and its scope ZCL is (ln(UB/LB) + 1)-competitive when value/weight ratios lie in [LB, UB] and items are small [6]. That covers estimated costs only. Binning, rollover and the cost estimator are heuristics.

4 Bins of S queries, budget B/N each, unused budget rolls over



Filled = spent. Outline = bin budget plus rollover. Arrows = carried-over budget.

Ours: S = 100 · LB = 0.5×min, UB = 2×max score/cost ratio on tuning data · affordability guard + explicit stop

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[6] Y. Zhou, D. Chakrabarty, and R. Lukose, “Budget constrained bidding in keyword auctions and online knapsack problems,” in Proc. 17th Int. Conf. World Wide Web, 2008, pp. 1243–1244.

A closer prior shrinks the regret bound, conditionally

Setting $r_t = \theta^* \cdot x_t + \varepsilon_t$, ε_t is R-sub-Gaussian

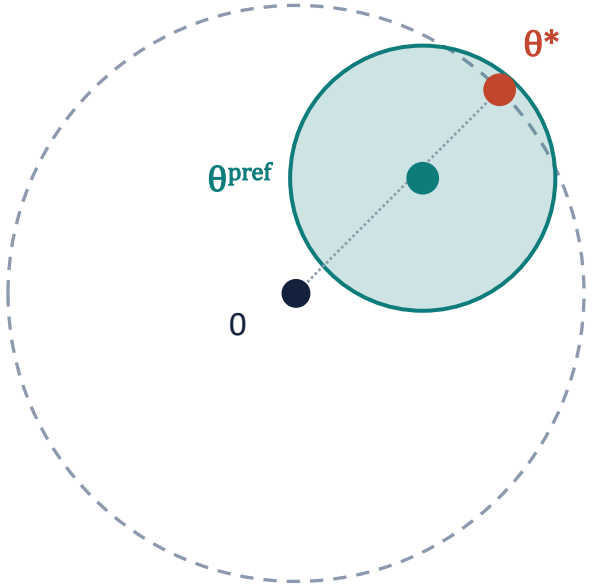
OFUL bound if $\|\theta^*\| \le S$: $R(T) \le U_T(S)$ w.p. $\ge 1 - \delta$

$$U_T(S) = 4\sqrt{(T\, d \log(\lambda + T/d))} \cdot \left(\sqrt{\lambda}\, S + R\sqrt{(2\log(1/\delta) + d \log(1 + T/(\lambda d)))} \right)$$

PI-OFUL start at θ^{pref}; bound $U_T(S')$ with $S' \ge \|\theta^* - \theta^{pref}\|$

Result $\|\theta^* - \theta^{pref}\| \le \|\theta^*\| \Rightarrow U_T(\text{PI-OFUL}) \le U_T(\text{OFUL})$

Intuition a good prior moves the centre of the confidence ball next to θ*, so the algorithm pays for less exploration.



Grey ball: radius $\|\theta^*\|$ around 0. Teal ball: radius $\|\theta^* - \theta^{pref}\|$ around the prior.

Limit 1 Same $d\sqrt{T}$ rate. Only the $\sqrt{\lambda} \cdot S$ constant shrinks.

Limit 2 S′ must be known. PILOT substitutes heuristics.

Limit 3 Proven for linear OFUL, not cosine-scored LinUCB.

Limit 4 Compares upper bounds, not realized regret.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[7] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, "Improved algorithms for linear stochastic bandits," in Adv. Neural Inf. Process. Syst., vol. 24, 2011, pp. 2312–2320.

Proof sketch, and two printing slips in the final PDF

1 · Ridge with a prior

$$A = X^\top X + \lambda I$$
$$b = X^\top y + \lambda \theta^{\text{pref}}$$

2 · Split the error

$$\tilde{\theta} - \theta^* = A^{-1} X^\top \varepsilon - \lambda A^{-1}(\theta^* - \theta^{\text{pref}})$$

3 · Confidence width

$$|x \cdot (\tilde{\theta} - \theta^*)| \le \|x\|_{A^{-1}} \times \left(R\sqrt{2 \log(\ldots/\delta)} + \sqrt{\lambda}\|\theta^* - \theta^{\text{pref}}\| \right)$$

4 · OFUL argument

same proof as OFUL
with S' in place of $S \Rightarrow U_T(S')$

Two printing slips (final ACL PDF)

p. 23945 b_t printed as $\lambda\theta^{\text{pref}} + \Sigma_s x_s$ — the reward r_s is missing.

p. 23946 bias printed as $\lambda(X^\top X + \lambda I)(\theta^* - \theta^{\text{pref}})$ — the inverse is missing.

With both restored, the ridge identity in step 2 holds. These are typographical slips, not a broken theorem.

Not covered by the proof

- Cosine normalization used by PILOT
- λ_a = 1/accuracy (accuracy is not a distance to θ^*)
- Cost policy, bins and rollover

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[7] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, “Improved algorithms for linear stochastic bandits,” in Adv. Neural Inf. Process. Syst., vol. 24, 2011, pp. 2312–2320.

Complexity against the predecessors

Method	Supervision needed	Routing time / query	Memory	Learning guarantee
LinUCB [2]	Reward of the chosen arm	$O(K\,d^2)$ with kept inverses	$O(K\,d^2)$	None for LinUCB itself. OFUL: $\approx d\sqrt{T}$
Hybrid LLM [3]	Both models' answers + quality scores, every training query	One transformer forward pass	Classifier weights	Supervised, no regret notion
PILOT, paper [1]	Chosen-arm reward + offline preference pairs	$\text{embed} + O(dD) + O(K\,d^2) + O(K)$	$O(K\,d^2 + dD)$	PI-OFUL: same $d\sqrt{T}$ rate, smaller constant if prior is close
Our implementation	Same as paper	$O(K\,d^3)$ dense solves	$O(K\,d^2)$: ≈ 260 KiB at $K = 8$, $d = 64$	Same as paper

0.07–0.24 s PILOT routing time per query in the paper

≈ 2.5 s one GPT-4 answer on MMLU, same paper

10–38× routing is that much faster than answering

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[2] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in Proc. 19th Int. Conf. World Wide Web, 2010, pp. 661–670.
[3] D. Ding et al., "Hybrid LLM: Cost-efficient and quality-aware query routing," in Proc. Int. Conf. Learn. Represent. (ICLR), 2024.

Core scope on a declared local variant

Our choice: core scope

E1 Performance vs budget, with the headline comparison

E2 Learning size and cumulative regret

+ Team-designed: 5 ablations, a budget-controller diagnosis, 2 critique tests on the poster

Supplement Local versions of six extra studies, labelled as such

Not full scope 3 of 11 models cannot be matched to RouterBench entries with evidence. No official code was found.

1,098
OpenAI conditions (858 + 240)

3
seeds

0
new paid requests

	Paper [1]	Ours
Model pool	11 RouterBench models	8 admitted. Claude-v2, Mistral-7B and Mixtral are unmatched.
Embedder	text-embedding-3-small (API)	Same model, real 1,536-d vectors → 64-d. Historical: BGE-M3 [9]
Data	RouterBench [8]: 36,497 queries, 64 tasks	Same 36,497 queries; 86 raw task labels
Splits	1,000 tuning; learning:deployment = 10:1	1,000 / 32,270 / 3,227 (MMLU 398 / 12,369 / 1,275)
Preferences	Arena subset; 500-pair test	1,245 train / 313 calibration (+ separate 847 / 211 / 500 supplement)
Seeds	not stated	7, 42, 2026: mean with min–max
Budget axis	fraction of GPT-4's cost	allocation of tuning-estimated GPT-4 cost; quality and cost ÷ same-query GPT-4

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[8] Q. J. Hu et al., "RouterBench: A benchmark for multi-LLM routing system," arXiv:2403.12031, 2024.
[9] J. Chen et al., "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," arXiv:2402.03216, 2024.

What the learner may see, and how we score it

RouterBench stream (to scale)



tuning 1,000 learning 32,270 deployment 3,227

Stage	Sets hyper-parameters	Sees chosen reward	Sees all rewards	Router updates
Tuning	α, window, cost	—	—	—
Learning	—	yes	—	chosen arm
Deployment	—	—	—	frozen
Evaluator only	—	yes	regret + quality	—

Same cost policy for every method. Tuned on OpenAI vectors: PILOT α = 1, LinUCB α = 5, EpochGreedy window 10. Seeds 7 / 42 / 2026.

Metrics

Quality / GPT-4

Σ reward ÷ GPT-4's Σ reward on the same requested queries. Unserved = 0.

Recorded cost / GPT-4

historical cost of the chosen answers ÷ GPT-4's cost on the same queries

Allocation

budget as a share of the tuning-estimated GPT-4 cost (50% ≈ 49% of GPT-4's recorded cost)

Coverage

served ÷ requested queries

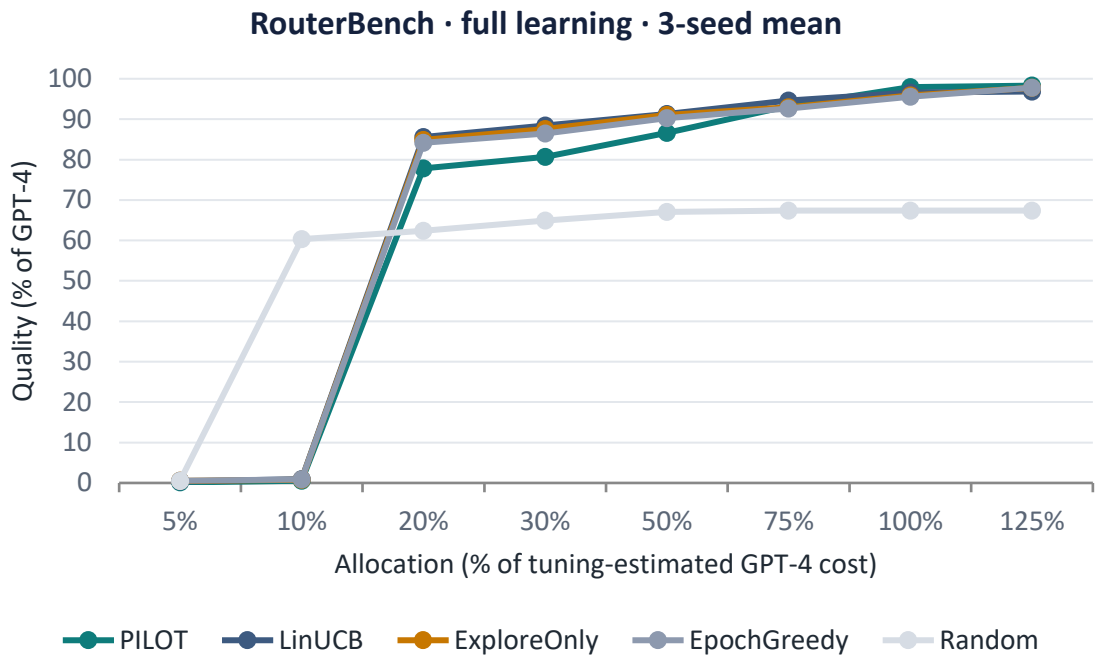
Oracle regret

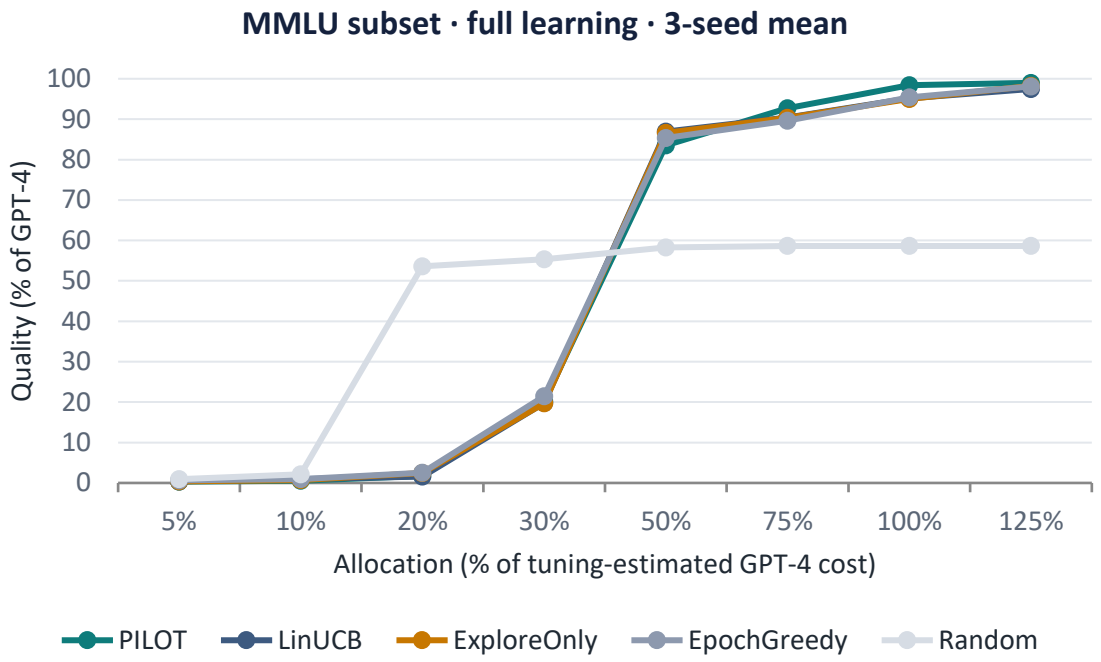
Σ over learning (best arm reward − chosen reward)

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.

RESULT · E1 PERFORMANCE VS BUDGET

PILOT trails at half allocation, leads only when money is loose





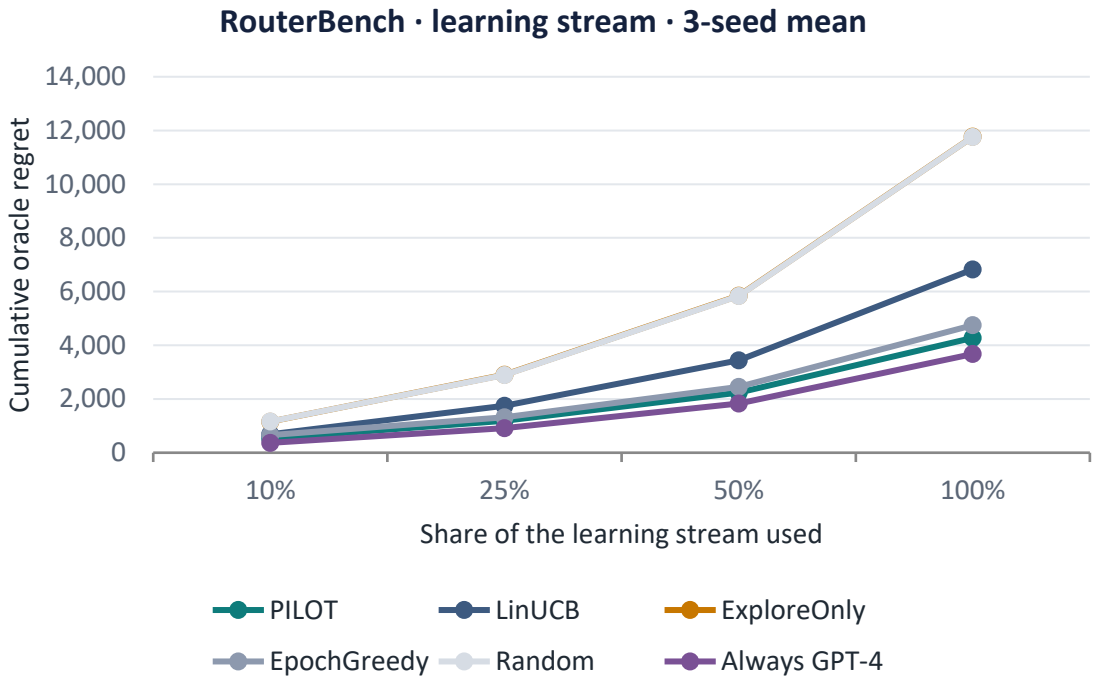
50% **PILOT 86.60%** at 46.23% cost · LinUCB 91.28% at 29.62%
100% **PILOT 97.94%** at 91.57% cost · LinUCB 96.59% at 49.44%
W&B PILOT 50%: s7 · s42 · s2026 LinUCB 50%: s7 · s42 · s2026

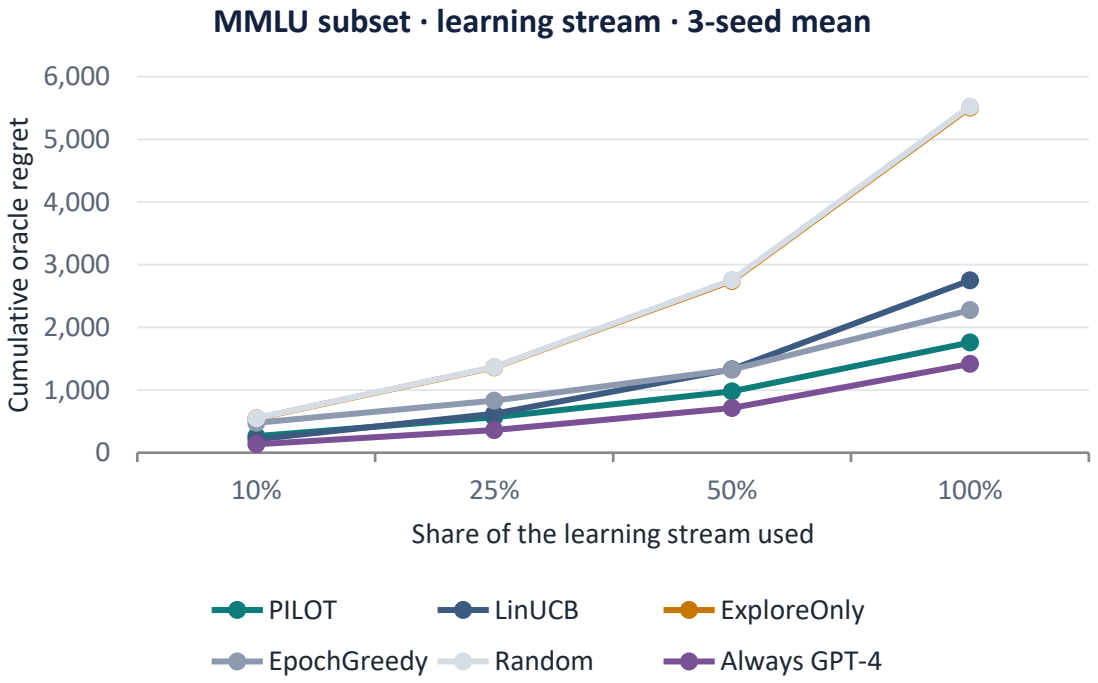
50% **PILOT 83.53%** at 47.53% cost · LinUCB 86.96% at 45.07%
100% **PILOT 98.44%** at 91.81% cost · EpochGreedy 95.43% at 72.66%
W&B PILOT 50%: s7 · s42 · s2026 LinUCB 50%: s7 · s42 · s2026

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

RESULT · E2 CUMULATIVE REGRET

Among adaptive learners, PILOT loses the least reward





4,275

vs 4,747 EpochGreedy, 6,826 LinUCB
always-GPT-4: 3,675 (not a learner)
Source: learning-prefixes.csv

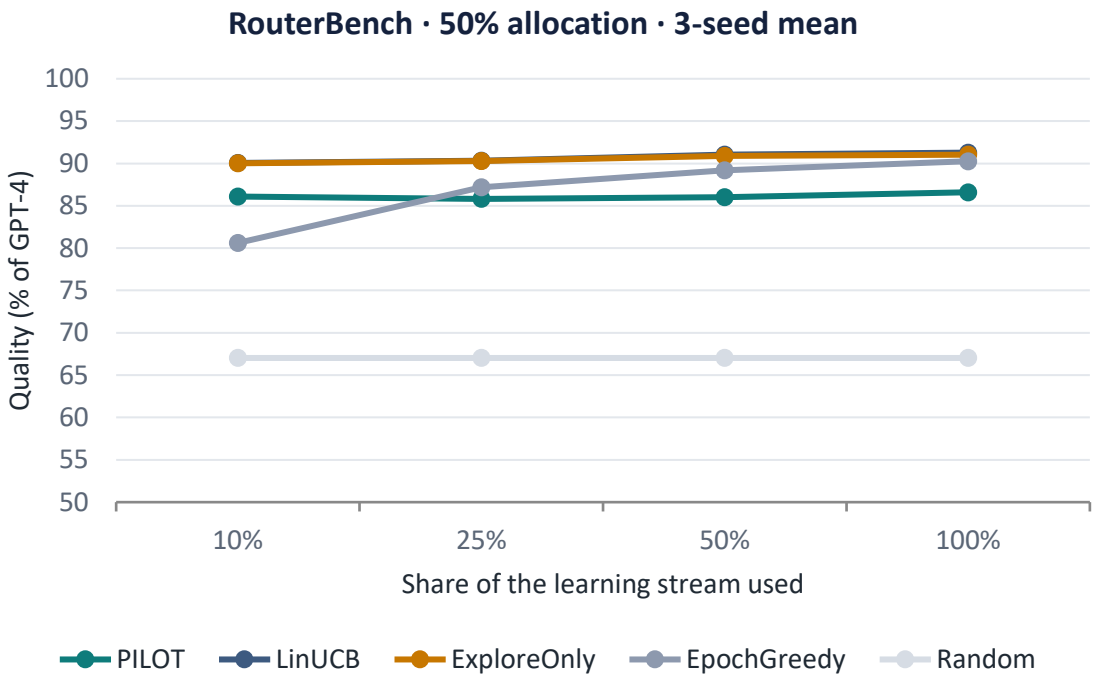
1,758

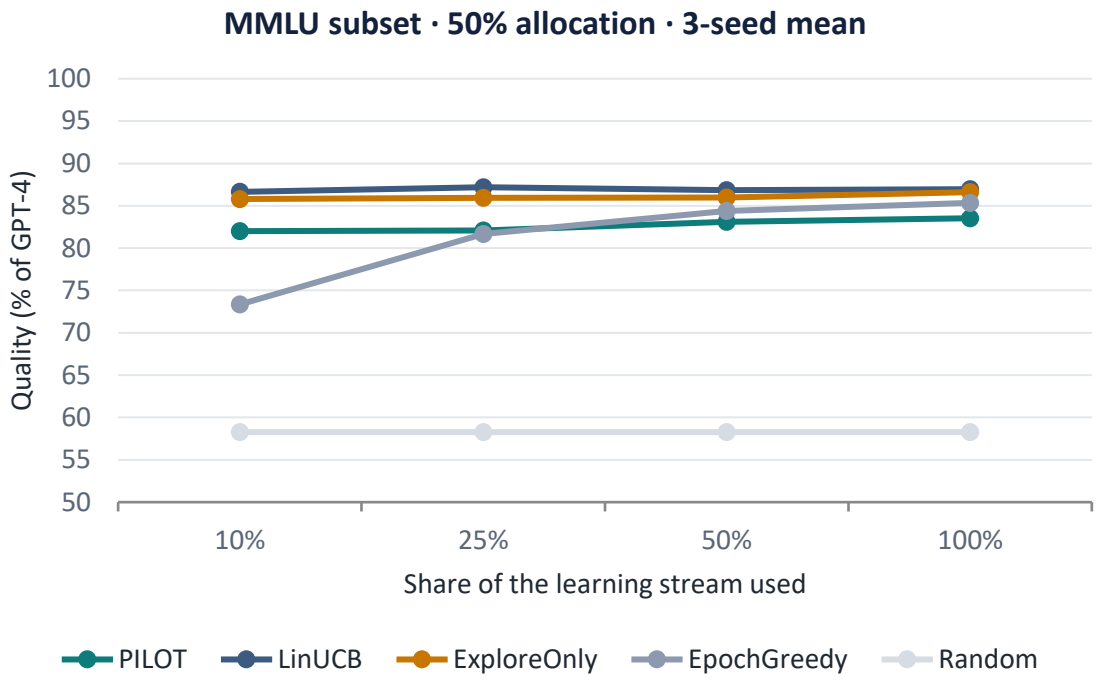
vs 2,274 EpochGreedy, 2,749 LinUCB
always-GPT-4: 1,416 (not a learner)
Source: learning-prefixes.csv

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

RESULT · E2 LEARNING SIZE

More learning barely moves PILOT at half allocation



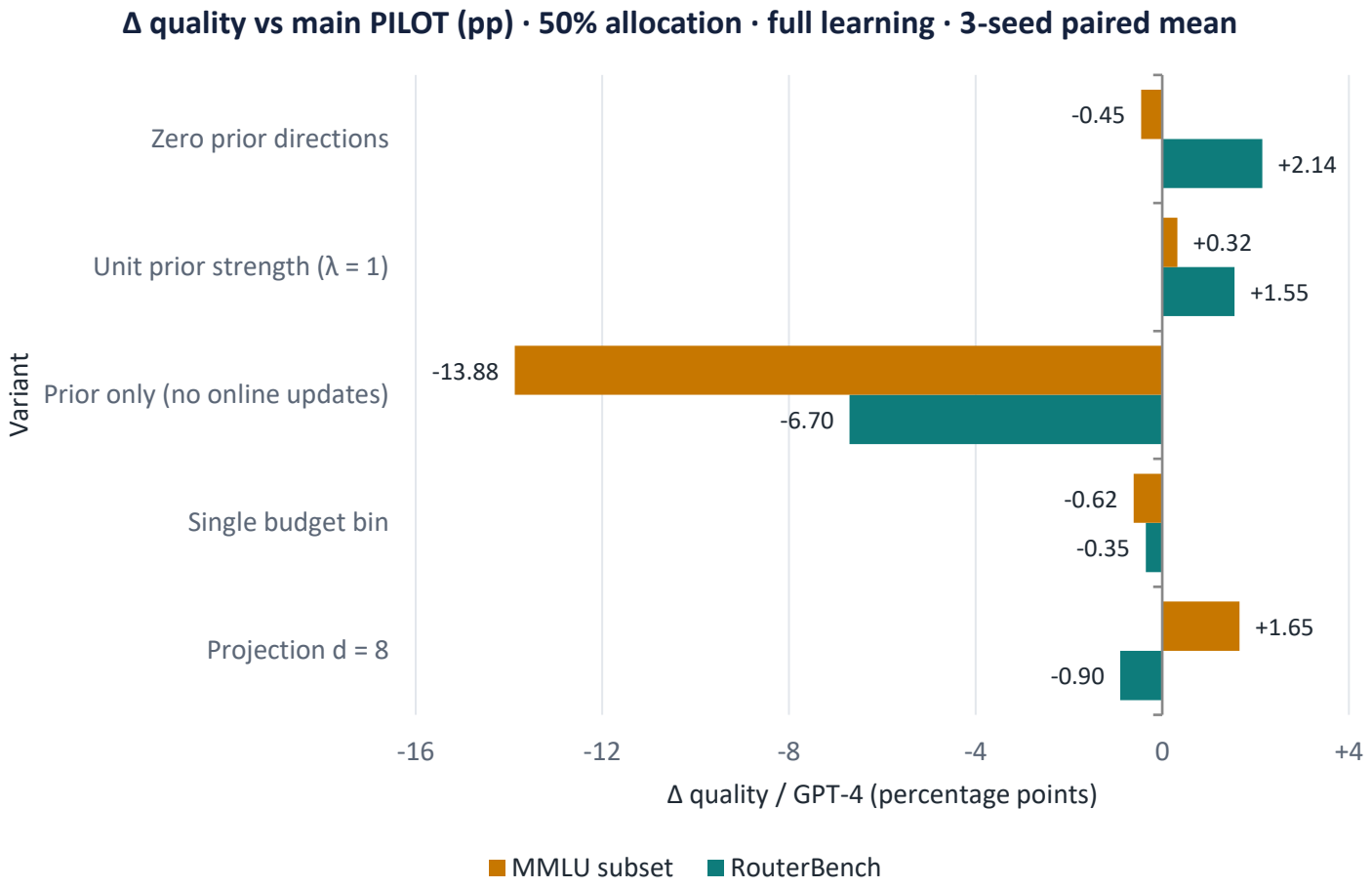


Lower regret ≠ better budgeted deployment. From 10% to 100% of the stream PILOT changes by +0.49 pp / +1.52 pp (RouterBench / MMLU), EpochGreedy by +9.65 pp / +12.00 pp. LinUCB stays highest at every prefix.

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

RESULT · TEAM-DESIGNED ABLATIONS

Online updates matter most; prior effects are mixed



Prior only
−6.70 pp / −13.88 pp: without online updates quality drops the most.

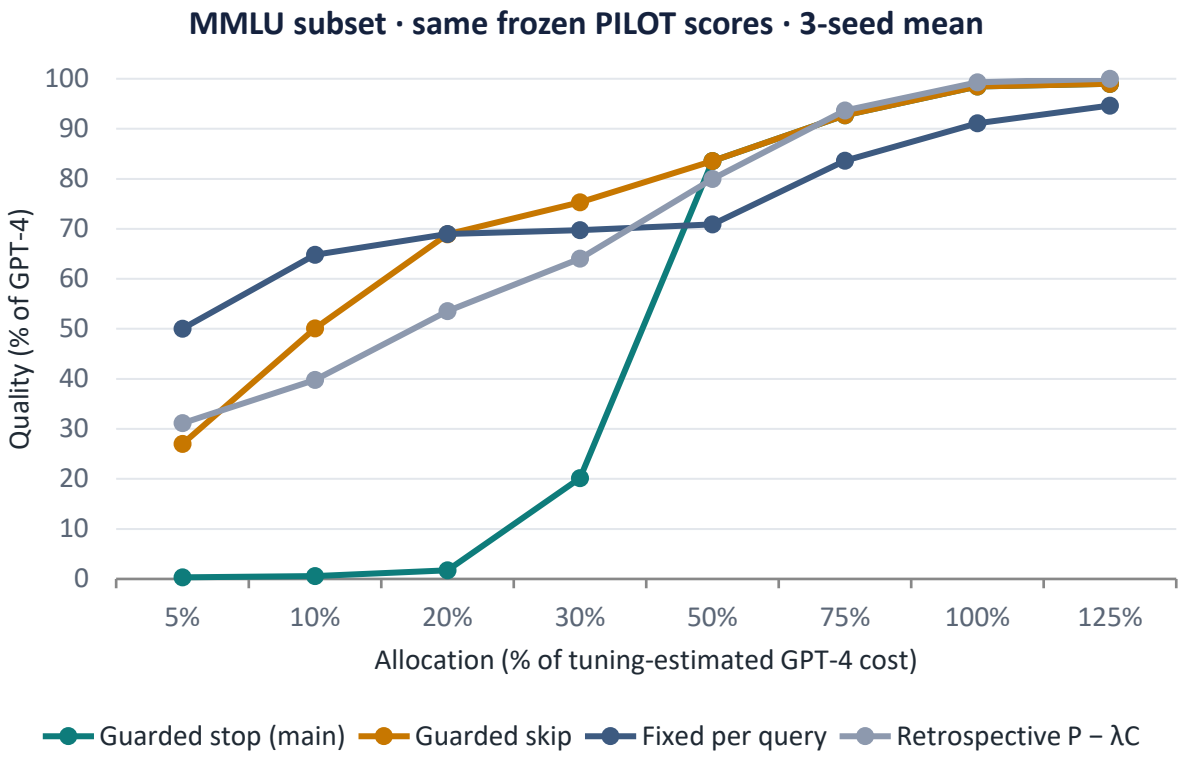
Zero prior directions
+2.14 pp / −0.45 pp: mixed sign. Projection and λ still come from preferences.

Unit strength λ = 1
+1.55 pp / +0.32 pp: the 1/accuracy heuristic is not supported here.

One bin, 8-d projection
−0.35 pp / −0.62 pp; −0.90 pp / +1.65 pp: small or mixed. One bin does not fix early stopping.

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

At tight allocations the stream stops while money remains



What happens at 20% on MMLU (all three seeds)

1. The bin is nearly spent, so the threshold admits no arm.
2. The fallback allowance (bin money ÷ queries left in the bin) is below the cheapest estimate.
3. Algorithm 3 stops: 29 of 1,275 queries served, ≥ 94% of the estimated budget unspent.

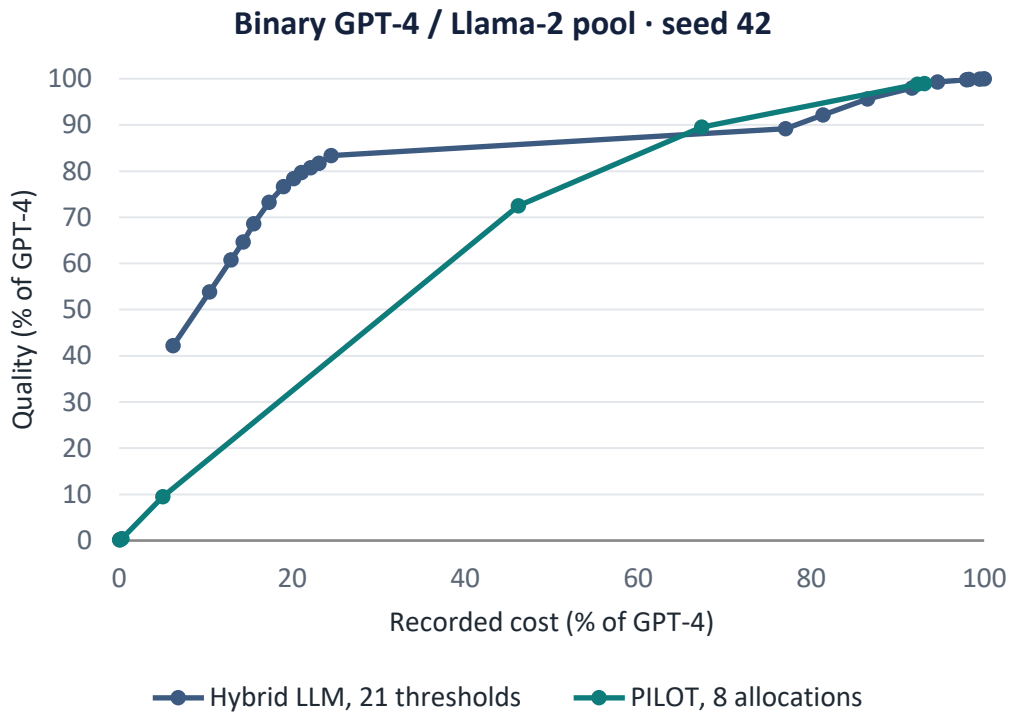
Alloc.	Policy	Quality / GPT-4	Coverage
20%	**Guarded stop (main)**	1.72%	2.27%
20%	**Guarded skip**	68.90%	97.31%
20%	**Fixed per query**	68.94%	99.92%
30%	**Guarded stop (main)**	20.17%	27.45%
30%	**Guarded skip**	75.29%	99.87%

W&B skip 20%: s7 · s42 · s2026
main 20%: s7 · s42 · s2026

Local extension, not a replacement. Same frozen scores, estimates and order; 240 replayed conditions. Paper-literal stop (no guard) equals the main controller everywhere; one bin is not a cure (15.18% at 20%). The retrospective comparator sees future predicted scores and estimated costs, not true rewards.

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

Local versions of the paper's extra studies give mixed results



Hybrid LLM [3] at threshold 0.5: **81.67% at 23.12% cost** · PILOT at 50% allocation: **72.49% at 46.16%**. Different controls, not a matched budget; Hybrid trains on both arms' rewards.

Paper study [1]	Our local result	Reading
Binary bandits App. D.4 · 3 seeds	50% alloc.: PILOT 72.45%, LinUCB 72.09%, EpochGreedy 70.81%. Regret 879 vs 729 / 731 (two-arm oracle).	**No universal PILOT win**
Task shift MMLU→GSM8K · seed 42	Before: PILOT 80.52% vs static Hybrid 79.58%. After: 65.61% vs 65.81% reward/query.	**No post-shift win**
Embedder [13] Sec. 5.4 · Instructor-XL	vs OpenAI: +2.58 pp / +0.55 pp at 50%, −0.60 pp / −0.23 pp at 100% (RB / MMLU).	**Small; recipe-dependent**
Preference prediction App. D.3 · local 500 pairs	RouteLLM-MF [12] 68.07% vs PILOT 63.53% (3 seeds, same pairs).	**Paper's ranking reversed**
Feedback noise App. D.5 · MMLU	0 / 1 / 3 / 5% replaced: 68.86 / 68.76 / 68.47 / 68.55% reward, 100% coverage.	**Mild**

Declared local adaptations: 8-arm or verified two-arm pools, our splits, one seed for neural baselines. Not exact author reproductions.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[3] D. Ding et al., "Hybrid LLM: Cost-efficient and quality-aware query routing," in Proc. Int. Conf. Learn. Represent. (ICLR), 2024.
[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[12] I. Ong et al., "RouteLLM: Learning to route LLMs with preference data," arXiv:2406.18665, 2024.
[13] H. Su et al., "One embedder, any task: Instruction-finetuned text embeddings," in Findings Assoc. Comput. Linguistics: ACL 2023, 2023, pp. 1102–1121.

RESULT · COMPARISON WITH THE PAPER

Paper and reproduction side by side

Claim or number	Paper [1]	Ours [10]	Deviation	Verdict
RouterBench headline	93% of GPT-4 quality at 25% of its cost	77.81% at 19.23% cost (20% alloc.) · 80.68% at 28.57% cost (30%)	not computed: no point at 25%, 8 vs 11 arms	not matched
MMLU headline	86% at 27%	stops early at 20–30% (coverage 2.27% / 27.45%) · 83.53% at 47.53% cost (50%)	not computed	not matched
PILOT vs best bandit baseline	best across budgets	50%: −4.68 pp RB, −3.44 pp MMLU · 100%: +1.35 pp RB, +3.02 pp MMLU	ranking flips with allocation	partial
Cumulative regret	lowest of all methods	lowest among learners: 4,275 vs 4,747 (RB), 1,758 vs 2,274 (MMLU); always-GPT-4 lower	same order among learners	reproduced (learners)
Preference prediction (App. D.3)	PILOT 65.0%, RouteLLM 63.6% (500 pairs)	local 500 pairs: PILOT 63.53%, RouteLLM-MF [12] 68.07%	different pairs and pool; no matched error	ranking reversed
Embedder (Sec. 5.4)	works with Instructor-XL	Instructor-XL [13] vs OpenAI: +2.58 pp / +0.55 pp at 50%	recipe comparison, retuned	consistent
Dataset	36,497 queries · 64 tasks · 11 models	36,497 queries · 86 raw labels · 8 models	pool −3 models	declared variant

Per-seed runs: Appendix and the follow-up experiment-map.csv. Calibration priors: evidence/priors.

[1] P. Panda et al., “Adaptive LLM routing under budget constraints,” in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[10] Team 8, “pilot-llm-routing-team8,” Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, “pilot-llm-routing,” GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing
[12] I. Ong et al., “RouteLLM: Learning to route LLMs with preference data,” arXiv:2406.18665, 2024.
[13] H. Su et al., “One embedder, any task: Instruction-finetuned text embeddings,” in Findings Assoc. Comput. Linguistics: ACL 2023, 2023, pp. 1102–1121.

RESULT · DEVIATION ANALYSIS

Why our numbers differ: causes and evidence

Gap	Candidate causes	Evidence from our runs	Status
PILOT trails at 20–75% (RB)	8-arm pool; 3 unverified version mappings; 86 vs 64 task labels; author split and settings	Gap persists with the paper's embedder. OpenAI − BGE [9]: −0.33 pp RB / −2.76 pp MMLU	**embedder ruled out; rest untestable**
Preference 60.70% vs 65.0%	313 calibration pairs vs a 500-pair test; different pairs and pool	Local 500-pair test: 63.53%. Zero directions: +2.14 pp / −0.45 pp	**not comparable**
λ heuristic	λ_a = 1/accuracy vs constant	λ = 1: +1.55 pp / +0.32 pp	**tested: not supported**
Collapse at low allocation	Explicit stop rule under binning	MMLU 20%: stop after 29 / 1,275; skip serves 97.31%	**explained**
Settings we fixed	S = 100, LB/UB, d = 64, margin, epochs, lr, guard	One bin −0.35 pp / −0.62 pp; d = 8 −0.90 pp / +1.65 pp; LB/UB untested	**partly tested**
Seed variation	3 seeds	50% RB: PILOT 84.98%–88.61% vs LinUCB 90.22%–91.93%	**ruled out**
Original code	No official implementation found (30 Oct 2026)	No divergence check possible	**not testable**

Libraries and hardware: outcomes are replayed and independent replays reproduce every learning group exactly, so floating-point differences cannot create these gaps.

[1] P. Panda et al., "Adaptive LLM routing under budget constraints," in Findings Assoc. Comput. Linguistics: EMNLP 2025, 2025, pp. 23934–23949.
[9] J. Chen et al., "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," arXiv:2402.03216, 2024.
[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8

TAKEAWAYS

What we can and cannot claim

Reproduced, with limits

- Among adaptive learners PILOT has the lowest learning regret on both populations; always-GPT-4 is lower.
- At loose allocations PILOT leads in quality (97.94% / 98.44% of GPT-4 at 100%) while spending about 92% of GPT-4's cost.
- Method, theory and complexity retold in our own notation.

Not reproduced locally

- At 50% allocation PILOT trails LinUCB (86.60% vs 91.28% RB, 83.53% vs 86.96% MMLU) and spends more.
- No matched 93%-at-25% point. The paper's embedder did not close the gap: OpenAI is not uniformly better than BGE.
- Locally RouteLLM-MF predicts preferences better (68.07% vs 63.53%).

Tested next (poster)

- Cosine scoring discards reward magnitude.
- Estimated-cost feasibility does not survive cost error.
- Both are argued theoretically and measured.

GitHub repository

W&B project
3,626 public runs

Scope: 8 arms, the paper's text-embedding-3-small (cached, no new paid request), three seeds, historical RouterBench outcomes. Not a matched replication of the 11-arm paper.

[10] Team 8, "pilot-llm-routing-team8," Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, "pilot-llm-routing," GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing

APPENDIX · TRACEABILITY

W&B runs behind every headline number

Number on the slides (quality / GPT-4)	Mean	Seed 7	Seed 42	Seed 2026
RouterBench PILOT 50%	**86.60%**	p2-7d78b1693ceca1bae751be80	p2-7e458ecb9aa6bc27b76bcaf3	p2-7325190ce97dc5c4a1d986fa
RouterBench LinUCB 50%	**91.28%**	p2-77db5e01f61381e958f9e169	p2-282b745642832ebb59f3d263	p2-1bc691d8055130960ac58419
MMLU PILOT 50%	**83.53%**	p2-e2a3a1f9474a22b0f9c5a31f	p2-7a1be896143dbf1dd1ea65cd	p2-ae43e5c37116d0d6205943b3
MMLU LinUCB 50%	**86.96%**	p2-e69603db58106be4ebffeb2e	p2-6e0469958272f93ee954d604	p2-647f1eb3313b84f07ae69543
RouterBench PILOT 100%	**97.94%**	p2-35f07caca3046f857c9467ae	p2-87e55a3701060f2df0175522	p2-3e032b4bce0bc6ad6f630001
RouterBench LinUCB 100%	**96.59%**	p2-8f35be049a2b04de12ce98e5	p2-335b9f4915abe7d60d423a70	p2-5b46026710d6ea299b560b94
MMLU PILOT 100%	**98.44%**	p2-53e75de1be2c2df31cbbbfba	p2-f76972412daa2e9f9b982419	p2-68cd995aaee7207a053c6910
MMLU EpochGreedy 100%	**95.43%**	p2-c318cec42a6256b9e553a9c3	p2-8a63f502544b1bf9ba055237	p2-e7411b1536a336cf3fb5c834

Every other point: the follow-up experiment-map.csv (2,498 rows: condition → source CSV → public run URL). Historical BGE claims: older map (1,128 rows). Regret: learning-prefixes.csv.

[10] Team 8, “pilot-llm-routing-team8,” Weights & Biases public project, 3,626 runs, Oct. 2026. [Online]. Available: forge.coreweave.com/wandb/kiettheis-university-of-science-vnu-hcm/pilot-llm-routing-team8
[11] Team 8, “pilot-llm-routing,” GitHub repository, Oct. 2026. [Online]. Available: github.com/kiettheis0702/pilot-llm-routing